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A DERIVATION OF THE WIENER RESPONSE

We expand the linear-MMSE argument of Sec. 3.2. Work in one channel and assume wide-sense stationarity, so that convolution diagonalizes in the Fourier basis and the mean-square error splits per frequency.

Setup. Under the mixture $x_t = a_t x_0 + b_t \varepsilon$ with ε white and independent of x_0 , let $S_x(f)$ be the power spectrum of x_0 . For a linear filter h_t with response $H_t(f)$, Parseval’s identity gives

$$J_t(h_t) = \sum_f \left(|H_t(f)|^2 (a_t^2 S_x(f) + b_t^2) - 2 \operatorname{Re}[H_t(f) a_t S_x(f)] + S_x(f) \right). \quad (16)$$

Optimality by differentiation. The summands decouple, so each $H_t(f)$ solves a scalar problem. Differentiating with respect to $\overline{H_t(f)}$ and setting to zero,

$$H_t(f) (a_t^2 S_x(f) + b_t^2) - a_t S_x(f) = 0, \quad (17)$$

which yields equation 7. The second derivative $a_t^2 S_x(f) + b_t^2 > 0$ confirms a minimum.

Power-law prior and spectral evolution. Natural images obey $S_x(f) \propto 1/|f|^p$ with $p \approx 2$. Inserting this into equation 7: at large t (small a_t) the denominator is noise-dominated except near $f = 0$, so G_t is low-pass; as $t \rightarrow 0$ ($a_t \rightarrow 1$) the passband widens toward high frequencies. Fig. 6 plots the exact responses along the flow schedule. This is the spectral evolution tracked by the filtered distance of Sec. 3.2, and the mean-gain normalization equation 8 makes those distances comparable across t : the raw gains in Fig. 6(a) differ by orders of magnitude across σ , whereas the normalized responses in (b) share a common scale and differ only in shape.

B PREDICTOR HIERARCHY

This section proves the three skip-length claims of Secs. 3.3–3.4: copy-reuse is at most linear in the skip length, degree- P Taylor remainder is of order $P+1$, and the Chebyshev truncation error does not depend on the skip length. Throughout, t_k is the last evaluated step and t_j is the skipped step at which the residual is reconstructed. The analysis is channel-wise, as in the main text. The order- P Taylor and Chebyshev truncation statements are those of Han et al. (2026, Thms. 3.1 and 3.2), restated here for the residual trajectory r in place of the hidden state; the copy-reuse bound and the node-spacing refinement of the discrete Taylor predictor are elementary.

Copy-reuse: linear bound. Copy-reuse sets $\hat{r}(t_j) = r(t_k)$. Assume r is L -Lipschitz on the sampling interval. Directly from the definition of Lipschitz continuity,

$$|r(t_j) - r(t_k)| \leq L |t_j - t_k|, \quad (18)$$

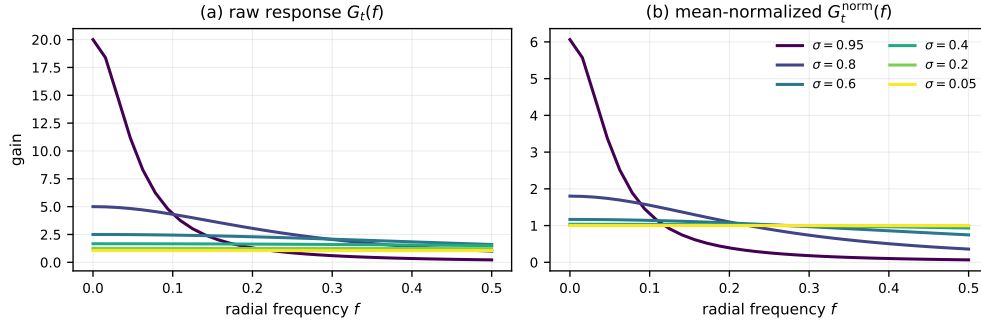


Figure 6: Wiener filter responses along the flow schedule ($a_t = 1 - \sigma$, $b_t = \sigma$, $S_x(f) = 1/|f|^p$, $p=2$ as used for FLUX; the video models use $p=3$ with the same qualitative shape). The formula is identical to SeaCache’s; only the plotted schedule instance differs (flow, as used by our gate, vs. their DPM illustration). (a) Raw response $G_t(f)$: low-pass early (large σ), widening as $\sigma \rightarrow 0$. (b) Mean-normalized $G_t^{\text{norm}}(f)$: comparable energy across timesteps, so accumulated distances are not dominated by gain drift. Computed from the closed form of equation 7.

which is equation 10. The bound is tight: $r(t) = Lt$ attains equality for every skip. Lipschitz continuity is also necessary in the worst case, since for a non-Lipschitz residual the ratio $|r(t_j) - r(t_k)|/|t_j - t_k|$ is unbounded as the skip shrinks. Copy-reuse uses no derivative information, so additional smoothness ($r \in C^P$ for $P \geq 1$) cannot improve this worst-case rate.

Taylor extrapolation: order- $(P+1)$ remainder. Assume $r \in C^{P+1}$. The degree- P Taylor expansion of r about t_k reads

$$r(t_j) = \sum_{p=0}^P \frac{r^{(p)}(t_k)}{p!} (t_j - t_k)^p + R_P(t_j), \quad (19)$$

with Lagrange remainder

$$R_P(t_j) = \frac{r^{(P+1)}(\xi)}{(P+1)!} (t_j - t_k)^{P+1} \quad (20)$$

for some ξ between t_k and t_j . Hence $|R_P(t_j)| \leq \frac{\|r^{(P+1)}\|_\infty}{(P+1)!} |t_j - t_k|^{P+1}$, a $(P+1)$ -order dependence on the skip length. This is the ideal predictor: it uses the exact derivatives at t_k , which a training-free cache never computes. In the discrete implementation the derivatives $r^{(p)}(t_k)$ are replaced by backward differences $\Delta^p r / (\Delta t)^p$ formed from $P+1$ neighboring evaluated residuals $s_0=t_k, s_1, \dots, s_P$, so the predictor is the degree- P interpolant p_P through those nodes, with remainder

$$r(t_j) - p_P(t_j) = \frac{r^{(P+1)}(\xi)}{(P+1)!} \prod_{q=0}^P (t_j - s_q) \quad (21)$$

for some ξ in the convex hull of $\{t_j, s_0, \dots, s_P\}$. Only the $q=0$ factor shrinks with the skip length $\ell = |t_j - t_k|$; the remaining factors are fixed by the positions of the earlier evaluated nodes. The $(P+1)$ -order rate therefore requires the node-spacing condition $\max_q |t_j - s_q| \leq C_{\text{node}} \ell$, under which $|r(t_j) - p_P(t_j)| \leq C_{\text{node}}^{P+1} \|r^{(P+1)}\|_\infty \ell^{P+1} / (P+1)!$. The dynamic schedule does not guarantee a uniform C_{node} : with the earlier nodes held fixed and $\ell \rightarrow 0$ the error is $O(\ell)$, not $O(\ell^{P+1})$. The failure is sharp, not a loose estimate. Take $P=1$, $r(t) = t^2$, nodes $s_0 = t_k = 0$ and $s_1 = -H$ with $H > 0$, and forecast point $t_j = \ell > 0$. The linear interpolant through $(0, 0)$ and $(-H, H^2)$ is $p_1(t) = -Ht$, so $|r(\ell) - p_1(\ell)| = \ell(\ell + H)$ and

$$\frac{|r(\ell) - p_1(\ell)|}{\ell^2} = 1 + \frac{H}{\ell}, \quad (22)$$

which is unbounded as $H/\ell \rightarrow \infty$: no uniform constant over arbitrary node configurations yields a bound of the form $|r(t_j) - p_P(t_j)| \leq C \ell^{P+1}$. Even uniform backward spacing does not recover

the ideal rate: for $t_j = t_k + \ell$ and $s_q = t_k - qh$ with $h > 0$,

$$\prod_{q=0}^P |t_j - s_q| = \prod_{q=0}^P (\ell + qh), \quad (23)$$

a degree- $(P+1)$ polynomial in ℓ that coincides with ℓ^{P+1} only in the limit $h/\ell \rightarrow 0$. In addition, the difference operator amplifies observation noise by binomial coefficients of order 2^P . A higher P therefore tightens the bound only for short skips and becomes unstable as $|t_j - t_k|$ grows. The predictor also uses only the $P+1$ most recent evaluations, discarding earlier trajectory information.

Chebyshev truncation: skip-length-independent bound. Map the solver index to $\tau \in [-1, 1]$ by an affine change of variable, and write r for the residual as a function of τ . Let E_ρ , $\rho > 1$, be the Bernstein ellipse with foci ± 1 and semi-axis sum ρ , and assume r extends analytically to E_ρ with $|r| \leq B$ there. The Chebyshev coefficients then admit the contour representation $c_m = \frac{1}{\pi i} \oint_{E_\rho} r(z) T_m(z)^{-1} dz/z$ up to the standard $m = 0$ factor. On E_ρ one has $|T_m| \geq (\rho^m - \rho^{-m})/2$, which yields $|c_m| \leq 2B\rho^{-m}$ for $m \geq 1$ and $|c_0| \leq B$. Since $|T_m(\tau)| \leq 1$ on $[-1, 1]$, truncating at degree M leaves

$$\|r - p_M^*\|_\infty \leq \sum_{m=M+1}^{\infty} 2B\rho^{-m} = \frac{2B}{\rho-1} \rho^{-M}, \quad (24)$$

which is equation 12. The right-hand side depends on M , ρ , and B , but not on the evaluation point τ_j . Consequently the approximation-theoretic term is independent of the skip length $|t_j - t_k|$. (The finite-sample ridge deviation from p_M^* is controlled separately by Lemma 1.)

Node spacing versus design conditioning. The relative node-spacing condition above and the design-conditioning quantity $\sigma_{\min}(\Phi)$ of Lemma 1 are distinct properties of the realized schedule: the former governs the local interpolation remainder, the latter the stability of the fitted Chebyshev coefficients, and neither implies the other. For one direction, let the forecast distance $\ell > 0$ stay fixed while the earlier nodes coalesce onto t_k ; then $\max_q |t_j - s_q|/\ell \rightarrow 1$, so the spacing condition holds with $C_{\text{node}} \rightarrow 1$, yet under the fixed affine normalization the rows of Φ become nearly identical and $\sigma_{\min}(\Phi) \rightarrow 0$ for any nonconstant basis. Conversely, a fixed well-spread K -point design keeps $\sigma_{\min}$ bounded away from zero while a forecast with $\ell \rightarrow 0^+$ against a fixed distant node drives $\max_q |t_j - s_q|/\ell \rightarrow \infty$. The ridge analysis of Lemma 1 is therefore not redundant alongside the interpolation analysis: each controls a separate failure mode.

C PROOF OF LEMMA 1 (RIDGE STABILITY)

The statement is Theorem 3.3 of Han et al. (2026). We reproduce the argument for the residual trajectory r in place of the hidden state, carrying the constants explicitly.

Fix one residual channel and drop the channel index, and take $\lambda > 0$. Write the K observations as $\mathbf{y} = \Phi \mathbf{c}^* + \mathbf{e}$, where $\mathbf{c}^*$ are the exact Chebyshev coefficients of p_M^* and, by equation 12, each entry of $\mathbf{e}$ satisfies $|e_k| \leq \varepsilon_M$, so $\|\mathbf{e}\|_2 \leq \sqrt{K} \varepsilon_M$. The ridge estimate is $\hat{\mathbf{c}} = (\Phi^\top \Phi + \lambda I)^{-1} \Phi^\top \mathbf{y}$, hence

$$\hat{\mathbf{c}} - \mathbf{c}^* = (\Phi^\top \Phi + \lambda I)^{-1} \Phi^\top \mathbf{e} - \lambda (\Phi^\top \Phi + \lambda I)^{-1} \mathbf{c}^*. \quad (25)$$

The forecast error at τ_j is the inner product of $\hat{\mathbf{c}} - \mathbf{c}^*$ with the basis row vector $\phi(\tau_j) = [T_0(\tau_j), \dots, T_M(\tau_j)]$, and $|T_m| \leq 1$ on $[-1, 1]$ gives $\|\phi(\tau_j)\|_2 \leq \sqrt{M+1}$. Let $\sigma_1 \geq \dots \geq \sigma_{M+1} = \sigma_{\min}$ be the singular values of $\Phi \in \mathbb{R}^{K \times (M+1)}$, with $\sigma_{\min} = 0$ whenever $K < M+1$. We bound the two terms of equation 25 in turn.

Variance term. In the singular-value decomposition of Φ ,

$$\|(\Phi^\top \Phi + \lambda I)^{-1} \Phi^\top\|_2 = \max_i \frac{\sigma_i}{\sigma_i^2 + \lambda} \leq \frac{\sigma_1}{\sigma_{\min}^2 + \lambda}, \quad (26)$$

and every entry of Φ is bounded by one, so $\sigma_1 \leq \|\Phi\|_F \leq \sqrt{(M+1)K}$. Combining this with $\|\phi(\tau_j)\|_2 \leq \sqrt{M+1}$ and $\|\mathbf{e}\|_2 \leq \sqrt{K} \varepsilon_M$,

$$|\phi(\tau_j)(\Phi^\top \Phi + \lambda I)^{-1} \Phi^\top \mathbf{e}| \leq \sqrt{M+1} \cdot \frac{\sqrt{(M+1)K}}{\sigma_{\min}^2 + \lambda} \cdot \sqrt{K} \varepsilon_M = \frac{(M+1)K}{\sigma_{\min}^2 + \lambda} \varepsilon_M. \quad (27)$$

Bias term. The Chebyshev coefficients of a function analytic on E_ρ and bounded by B there satisfy $|c_m^*| \leq 2B\rho^{-m}$ (App. B), so

$$\|\mathbf{c}^*\|_2 \leq 2B \left(\sum_{m \geq 0} \rho^{-2m} \right)^{1/2} = \frac{2B}{\sqrt{1 - \rho^{-2}}}, \quad (28)$$

a bound independent of M . Since the eigenvalues of $(\Phi^\top \Phi + \lambda I)^{-1}$ are $1/(\sigma_i^2 + \lambda)$, its spectral norm is $1/(\sigma_{\min}^2 + \lambda)$, and therefore

$$|\phi(\tau_j) \lambda (\Phi^\top \Phi + \lambda I)^{-1} \mathbf{c}^*| \leq \frac{\lambda \sqrt{M+1}}{\sigma_{\min}^2 + \lambda} \cdot \frac{2B}{\sqrt{1 - \rho^{-2}}}. \quad (29)$$

Combination. Adding the two bounds gives Lemma 1. Adding ε_M once more, by equation 12 and the triangle inequality, bounds the total error $|r(\tau_j) - \hat{r}(\tau_j)|$ and recovers Theorem 3.3 of Han et al. (2026). The two terms expose the trade-off: further observations raise $\sigma_{\min}$ and keep the variance factor bounded, while λ guards the rank-deficient regime $K < M+1$ at the price of bias.

D RESIDUAL-SPACE ERROR IDENTITY AND CONDITIONING

On a skipped step the block input $\mathbf{h}_{\text{in}}$ is known exactly from the current sampling state and the preceding computation, so with $\mathbf{h}_{\text{out}} = \mathbf{h}_{\text{in}} + \mathbf{r}$,

$$\hat{\mathbf{h}}_{\text{out}} - \mathbf{h}_{\text{out}} = \hat{\mathbf{r}} - \mathbf{r}, \quad (30)$$

i.e. state and residual forecasts share the same error for a fixed fit. The fits differ in their regression target. Stacking the observed input, output, and residual rows gives matrices satisfying the exact identity $\mathbf{H}_{\text{out}} = \mathbf{H}_{\text{in}} + \mathbf{R}$, where $\mathbf{H}_{\text{in}}$ varies with the sampling step and is not rank one, so no low-rank decomposition is assumed. Each hidden-state observation carries the large input component, which the sampler already knows exactly; fitting the full state therefore inflates the regression target without adding anything that needs predicting. The ridge map is linear in its target, so scaling the target scales the fitted coefficients and the absolute shrinkage bias while the shrinkage factor $\lambda/(\sigma_{\min}^2 + \lambda)$ is unchanged (App. C): the ridge constants scale with the target norm through B , which bounds the analytic continuation of the fitted trajectory. Forecasting $\mathbf{R}$ directly keeps the target small-normed and tightens every constant in proportion, a target-scale advantage rather than a design-conditioning one — all fits share the same design matrix Φ and hence identical conditioning. The skipped-step update is $\mathbf{h} \leftarrow \mathbf{h} + \hat{\mathbf{r}}$.

E BLEND OPTIMUM AND WARM-UP THRESHOLD

Blend weight. With uncorrelated Taylor error (squared bias b_T^2) and Chebyshev error (variance σ_C^2), the blended risk is $R(w) = (1-w)^2 b_T^2 + w^2 \sigma_C^2$. Setting $R'(w) = 0$ gives $w^* = b_T^2 / (b_T^2 + \sigma_C^2)$. Early in sampling both the drift and the fit variance are large; late in sampling both shrink as the trajectory settles and K grows (Appendix C). Measured trajectories keep the two terms within a factor of two of each other, which places w^* near $1/2$ throughout and motivates the fixed $w = 0.5$.

Warm-up. With n observations and $M+1$ bases, $n < M+1$ leaves equation 11 rank-deficient: $\sigma_{\min}(\Phi) = 0$, the observations cannot identify all coefficient directions, and the fit is dominated by the prior (the bias term of Appendix C governs). The first-order discrete Taylor step is exact on the last two observations by construction and therefore strictly dominates in this regime. Blending from $n \geq 4$ onward (the $M = 4$ fit being determined in practice) is thus the degrees-of-freedom crossover rather than a tuned choice.

F REFRESH-RATIO ANALYSIS

Refresh histograms concentrate compute early along the trajectory, in accordance with the spectral prior of equation 7: evaluations are dense while G_t is narrow (content forming) and sparse as the passband widens. PSNR-versus-refresh curves for ReSPect follow the full-compute reference monotonically at both δ settings, with the gap to copy-reuse widening at lower refresh ratios, precisely where the reuse bound equation 10 loosens and the uniform bound equation 12 remains tight, as anticipated in Sec. 4.2.

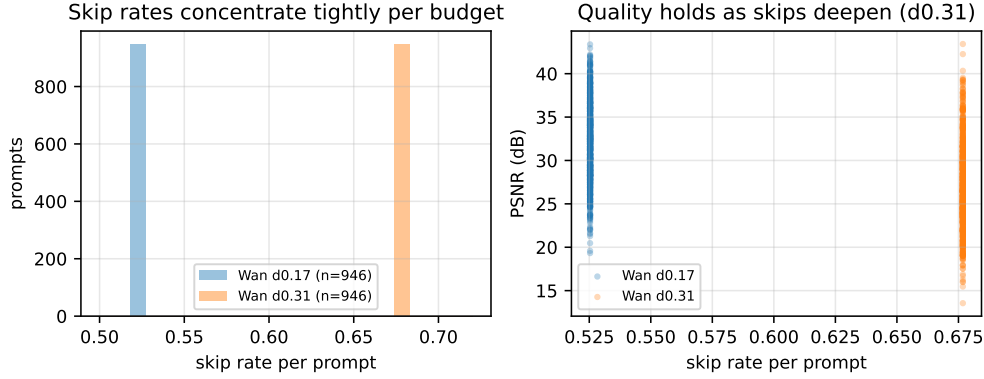


Figure 7: Compute concentrates early; quality tracks full compute. Left: per-prompt skip-rate histograms (Wan). Right: per-prompt PSNR vs skip rate over full VBench-946.

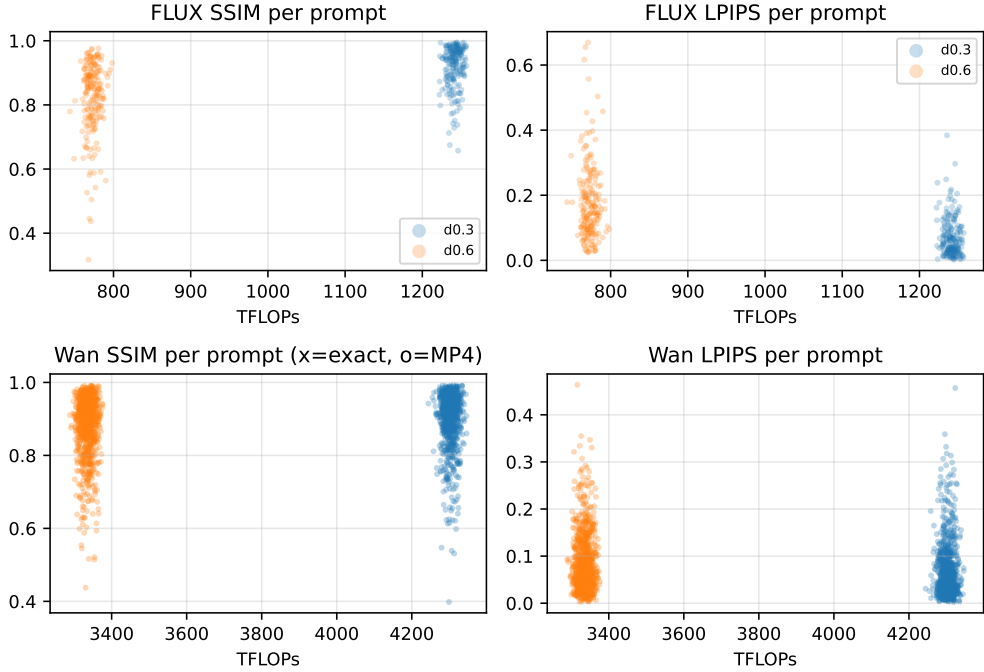


Figure 8: Per-prompt SSIM/LPIPS clouds at each config’s TFLOPs. Top: FLUX DrawBench-200. Bottom: Wan VBench-946 .

G IMPLEMENTATION DETAILS

Forecaster. $M=4$ Chebyshev bases, window $K=100$, $\lambda=0.1$, blend $w=0.5$, first-order discrete Taylor, Taylor-only warm-up until $n \geq 4$ observations. Time is mapped to $\tau \in [-1, 1]$ over the fixed range $[0, 50]$ (buffer min/max is *not* used). One forecaster per sample; coefficient state is a single $(M+1) \times F$ matrix plus K cached residuals.

Complexity. Fitting is $\mathcal{O}(K(M+1)F)$ with $M \ll K \ll F$ (the $(M+1)^3$ Cholesky term is negligible); prediction is $\mathcal{O}((M+1)F)$; memory is $\mathcal{O}((M+1)F + KF)$. Measured overhead is ~ 0.09 s/image on FLUX. First and last steps always compute; CFG call streams are demuxed by timestep equality.

Hardware and seeds. FLUX.1-dev, DrawBench (200 prompts), 1024×1024 , 50 steps, guidance 3.5, bf16, seeds $0+i$. Video: VBench prompts, 480p, 65 frames, 50 steps. All latencies are wall-clock on a single NVIDIA B200; all quality claims are at matched NFE.

Cost of the forecaster. A latency column combines the effects of schedule and reconstructor, since two methods at the same budget may skip at different steps. Running both reconstructors in one harness at the same δ , which fixes the schedule and varies only what is written on a skipped step, yields identical realized NFE: the gate reads the modulated block input, so replacing copy-reuse with a forecast does not alter where it fires. The forecast adds 1.7–4.3% wall-clock over copy-reuse (Tab. 5).

Table 5: Cost of the reconstructor at fixed schedule. Both columns of each row use the same gate and the same δ in the same harness on one B200; only the skipped-step reconstruction differs. NFE is identical within every pair.

Model	δ	NFE	Copy-reuse (s)	Forecast (s)	Δ
FLUX.1-dev	0.3	21	2.38	2.47	+3.8%
FLUX.1-dev	0.6	13	1.55	1.61	+3.9%
Wan2.1-1.3B	0.17	46	14.31	14.80	+3.4%
Wan2.1-1.3B	0.31	32	10.49	10.94	+4.3%
HunyuanVideo	0.27	24	39.20	39.86	+1.7%
HunyuanVideo	0.48	16	27.38	27.90	+1.9%

ReSpect vs. SeaCache at matched schedules. We further compare ReSpect against SeaCache (Chung et al., 2026) copy-reuse under matched schedules, tuning δ to equalize NFE. Both arms run in our own harness: the gate is unchanged, so the two methods realize (near-)identical NFE and only the skipped-step reconstruction differs. Tab. 6 reports full-reference quality against the uncached outputs. The forecast wins on PSNR, SSIM, and LPIPS at every setting with a matched pair: +0.79 dB on FLUX $\delta=0.37$ ($n=200$), +0.49 dB on FLUX $\delta=0.6$ ($n=20$), and +0.93 dB on Wan $\delta=0.31$ ($n=20$), with LPIPS down 18% where measured. We attribute this to the residual forecast tracking the trajectory between evaluations, whereas copy-reuse holds the last observed residual fixed.

Table 6: ReSpect vs. SeaCache copy-reuse at matched schedules: same δ , same prompts, (near-)identical NFE; only the skipped-step reconstruction differs. LPIPS was not measured on the 20-prompt subsets (—). The FLUX $\delta=0.37$ pair is full DrawBench-200; the $\delta=0.6$ and Wan pairs are fixed 20-prompt subsets.

Setting	Method	NFE	PSNR $\uparrow$	SSIM $\uparrow$
FLUX $\delta=0.37$ ($n=200$)	SeaCache	18.1	25.45	0.880
	ReSpect	18.1	26.24	0.895
FLUX $\delta=0.6$ ($n=20$)	SeaCache	13.0	23.08	0.848
	ReSpect	13.0	23.57	0.852
Wan $\delta=0.31$ ($n=20$)	SeaCache	32.4	28.80	0.904
	ReSpect	32.3	29.73	0.917

H RECONSTRUCTOR ABLATIONS AGAINST COPY-REUSE

All arms share one base run, so every Δ dB value below is paired against the same copy-reuse baseline (Fig. 9), run at the aggressive budgets (FLUX $\delta=0.6$, Wan $\delta=0.31$) on fixed 20-prompt subsets.

Blend weight w . Pure Taylor ($w=0$) is strong wherever neighboring evaluations carry the signal: +2.30 dB on Wan and +0.50 dB on FLUX. Pure Chebyshev ($w=1$) fails precisely where the fit is least determined: −1.68 dB on Wan, with its long skip runs, and +0.09 dB ($\approx$ copy-reuse) on FLUX. The fixed $w=0.5$ remains at or above copy-reuse on either model (+0.49/+0.93 dB); it is a robust compromise, and the setting used in the main tables, rather than a per-subset optimum.

Warm-up threshold. Blending before the fit is determined ($n < M+1$) lets the penalty dominate the data. A warm-up of 8 attains the largest FLUX gain (+0.68 dB) but lies within noise of the default blend; longer Taylor-only spans (12–24) never accumulate enough observations to blend at this budget and coincide with pure Taylor (+0.50 dB). Extending the warm-up is therefore without penalty, and the polynomial term engages only once the fit is determined.

Window K . $K=32$ matches $K=100$ to three decimals (+0.491 vs. +0.491 dB): per-prompt buffers rarely exceed ~ 40 observations, so the $\sigma_{\min}$ gains of Lemma 1 have saturated early and $K=100$ is a conservative default rather than a tuned choice.

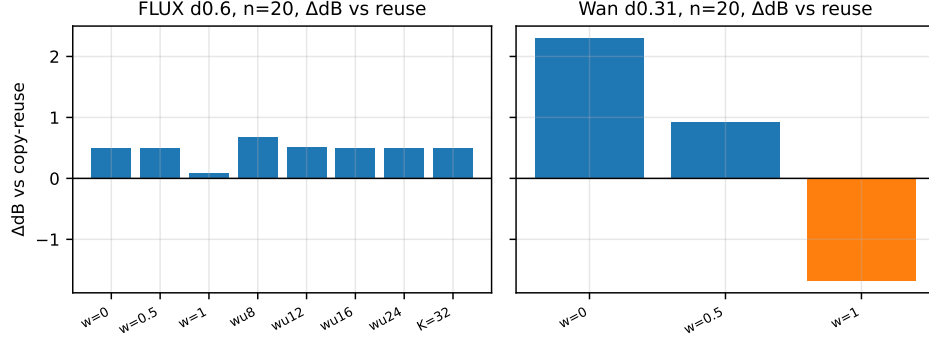


Figure 9: Ablations (Δ dB against copy-reuse, paired on one shared base run per panel). Left: FLUX $\delta=0.6$, $n=20$. Right: Wan $\delta=0.31$, $n=20$. Pure Taylor dominates on Wan while the blend remains at or above copy-reuse; pure Chebyshev fails on long skip runs; warm-ups of 12 and above never engage the blend; the window K has saturated.

I ADDITIONAL QUALITATIVE RESULTS

All clips below are generated in our harness on one B200. ReSPect runs at $\delta=0.31$ (Wan, 32 forward passes) and $\delta=0.48$ (Hunyuan, 16); TeaCache and TaylorSeer run at their published operating points and use more compute. The two sentence prompts are (retriever) “A golden retriever running along a sandy beach at sunset, waves rolling in behind it” and (balloon) “A red hot air balloon drifting slowly over green rolling hills under a clear blue sky”; both are also printed in the title of each strip figure. Per-frame PSNR against the base frame is printed under each panel. Averaged over all 65 frames, ReSPect leads on every clip: Wan balloon 34.87 vs. 31.29 / 17.60; Wan retriever 30.96 vs. 27.07 / 17.91; Hunyuan balloon 37.41 vs. 25.80 / 21.52; Hunyuan retriever 26.04 vs. 16.32 / 15.24 (TeaCache / TaylorSeer). The retriever prompt is a difficult motion case in which all caches degrade while the ordering is preserved. Tab. 7 gives the full per-prompt comparison with SSIM and LPIPS (frame-averaged against the uncached same-seed reference).

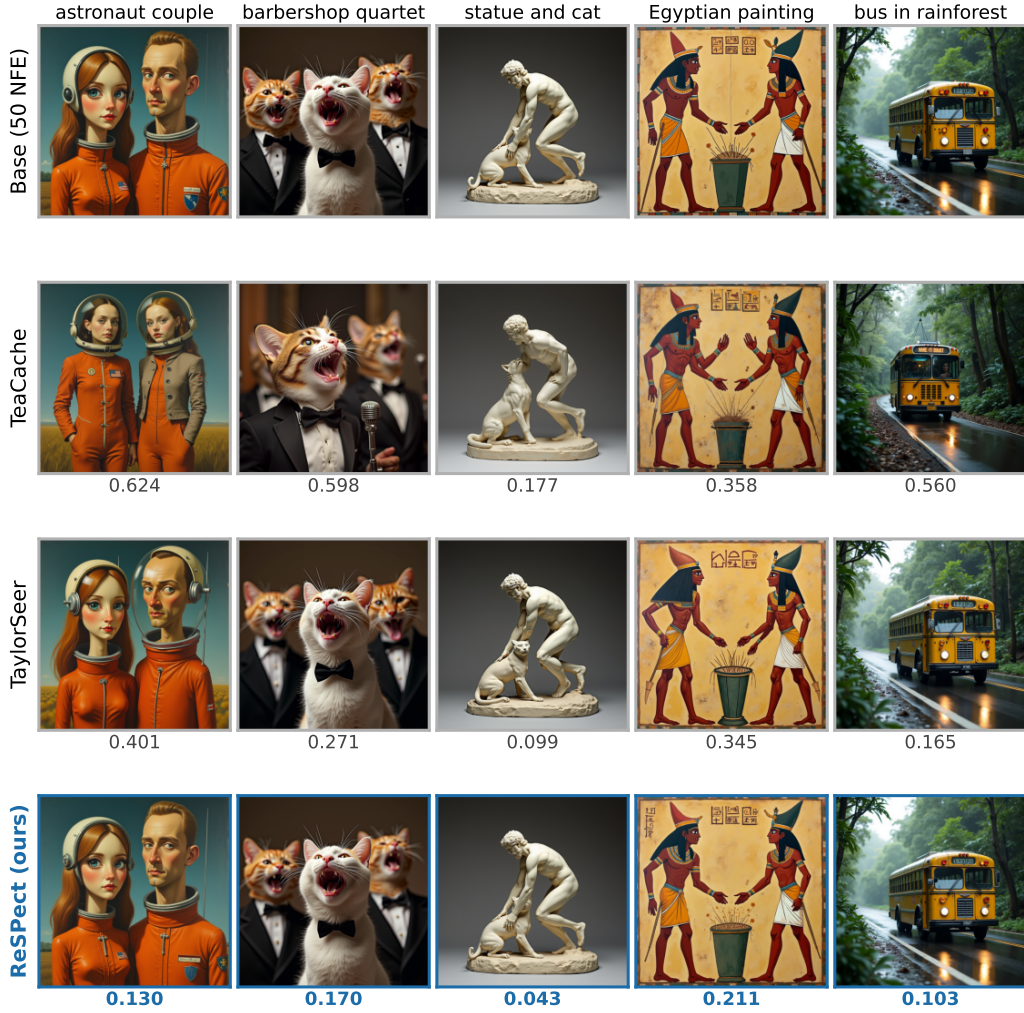


Figure 10: FLUX.1-dev at the aggressive budget, all methods run in our harness. ReSpect at $\delta=0.6$; TaylorSeer ($\mathcal{S}=5$) and TeaCache ($\delta=0.6$) use equal or more compute. LPIPS against the base image is printed under each panel. Over 20 DrawBench prompts ReSpect attains the lowest LPIPS on 19 (mean 0.183 vs. 0.256 and 0.407).

Table 7: Per-prompt qualitative-clip comparison: frame-averaged PSNR, SSIM, and LPIPS against the uncached same-seed reference over all 65 frames. NFE is forward passes per clip (Wan base 100, Hunyuan base 50). ReSpect uses the least compute and leads on every clip and metric.

Clip	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	NFE
Wan retriever	TeaCache	27.07	0.892	0.109	38
	TaylorSeer	17.91	0.656	0.384	36
	ReSpect	30.96	0.937	0.066	32
Wan balloon	TeaCache	31.29	0.918	0.064	38
	TaylorSeer	17.60	0.568	0.321	36
	ReSpect	34.87	0.950	0.031	32
Hunyuan retriever	TeaCache	16.32	0.669	0.322	18
	TaylorSeer	15.24	0.643	0.379	18
	ReSpect	26.04	0.878	0.089	16
Hunyuan balloon	TeaCache	25.80	0.914	0.046	18
	TaylorSeer	21.52	0.903	0.117	18
	ReSpect	37.41	0.980	0.005	16

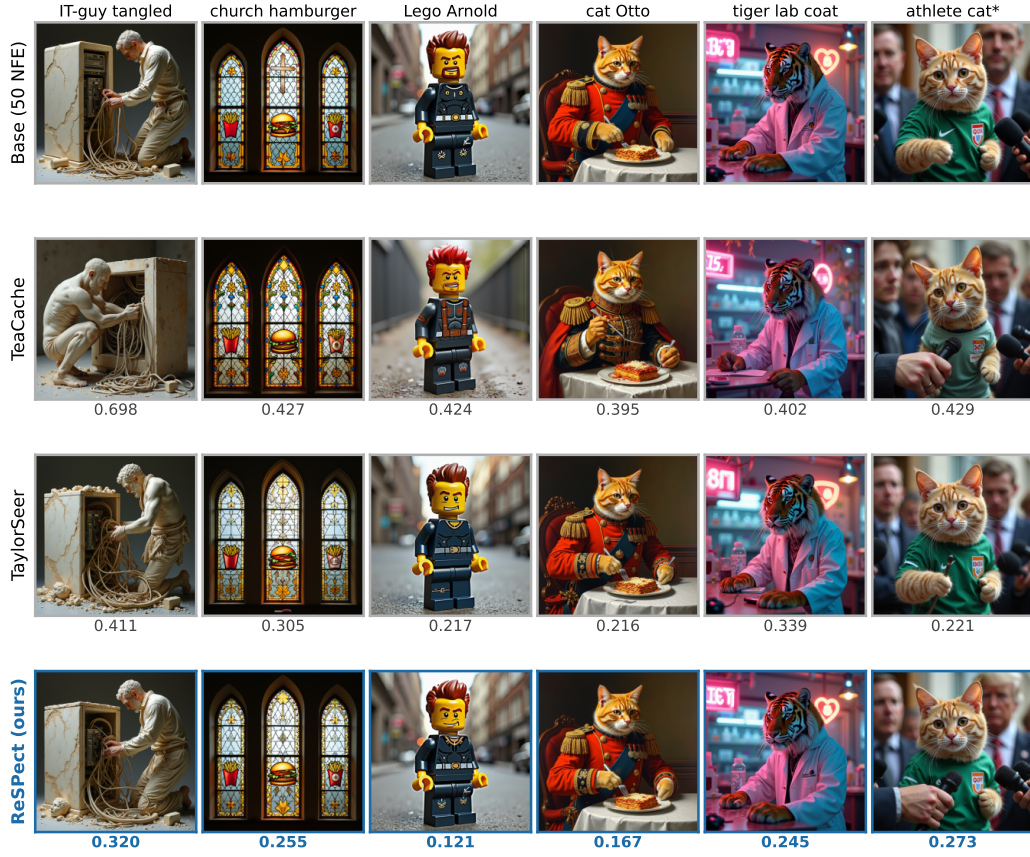


Figure 11: Six further DrawBench prompts at the same aggressive budget and harness as Fig. 10 (ReSPect $\delta=0.6$; TaylorSeer $S=5$; TeaCache $\delta=0.6$). ReSPect is closest to the base on five of six; “athlete cat*” is the single prompt of the twenty where TaylorSeer attains the lower LPIPS (0.221 vs. 0.273), shown here for completeness.

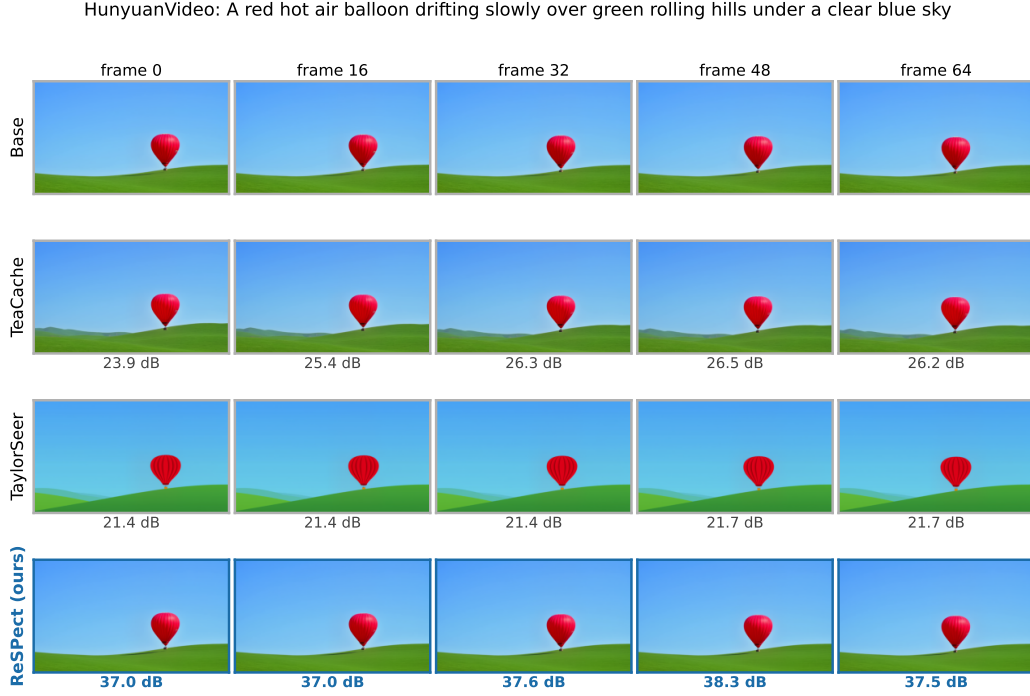


Figure 12: HunyuanVideo, balloon clip, $\delta=0.48$ (16 NFE). TaylorSeer renders a different scene entirely and retains it for the whole clip; averaged over all 65 frames, ReSPect scores 37.41 dB against 25.80 (TeaCache) and 21.52 (TaylorSeer).

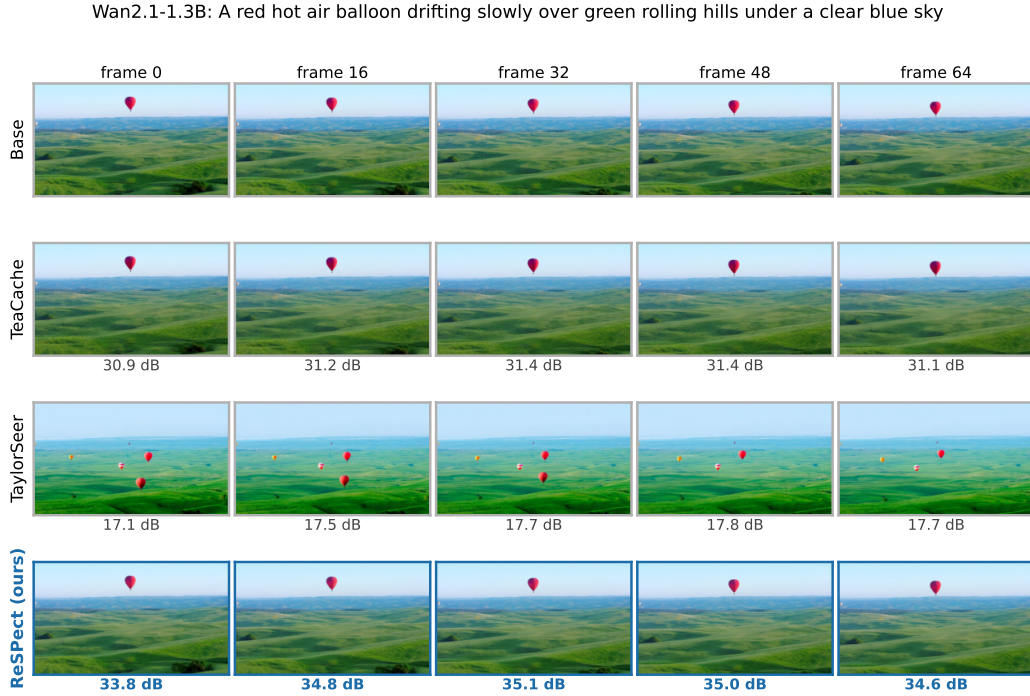


Figure 13: Wan2.1, balloon clip, $\delta=0.31$ (32 NFE). TaylorSeer hallucinates three additional balloons and retains them for the whole clip.

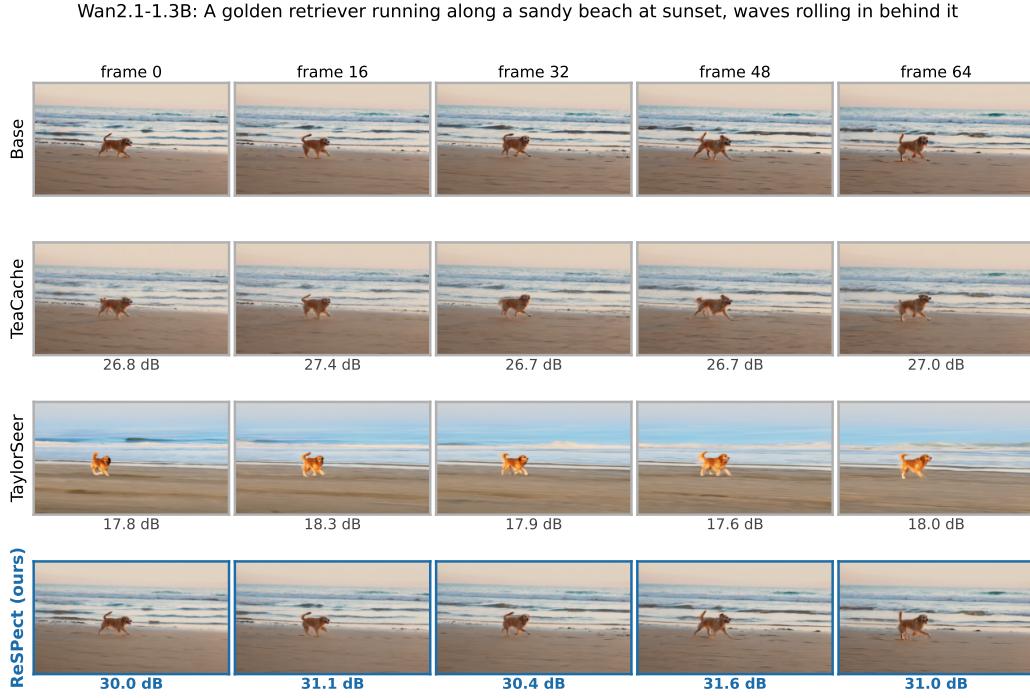


Figure 14: Wan2.1, retriever clip, $\delta=0.31$ (32 NFE): a difficult motion prompt. All methods lie closer to the base here; the forecast still gains +3.9 dB over the next baseline, whereas the interval-based and raw-distance schedules lose 4–13 dB.

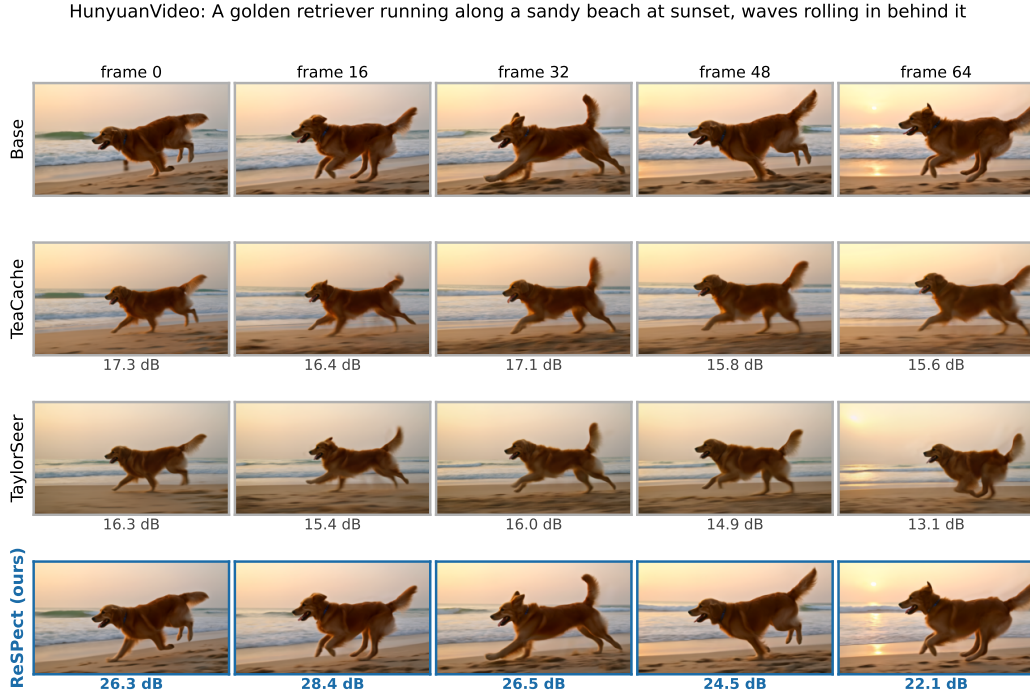


Figure 15: HunyuanVideo, retriever clip, $\delta=0.48$ (16 NFE).